

(TR-104) PYTHON WITH AI/ML

Training Day 63 Report

Date: 8 April 2026

Topic Covered: Revision of Previous Concepts (AI Systems, RAG, MCP, Agents & LLM Integration)

The sixty-third day of training focused on revising all previously covered concepts related to AI systems, workflow orchestration, LLM integration, and agent-based architectures. The session was dedicated to strengthening theoretical understanding and consolidating practical knowledge gained throughout the project.

- Revised core concepts of RAG (Retrieval-Augmented Generation) and its complete pipeline flow.
- Revisited MCP (Model Context Protocol) and its role in connecting LLMs with external tools and systems.
- Revised AI Agents architecture and how agents differ from traditional chatbot systems.
- Reviewed tool calling mechanisms and structured output handling in LLM-based applications.
- Revisited FastAPI backend integration with AI workflows and model endpoints.
- Revised multi-LLM integration approach including Ollama, Hugging Face, and Gemini routing.
- Strengthened understanding of guardrails including input validation, prompt injection detection, and output filtering.
- Reviewed workflow engine design including node-based execution and agentic loops.
- Revised frontend integration using Streamlit and React Flow for AI workflow visualization.
- Consolidated knowledge of vector databases, embeddings, and retrieval mechanisms used in RAG systems.
- Improved clarity on end-to-end AI system architecture combining LLMs, tools, APIs, and workflows.

- Strengthened overall understanding of production-level AI system design and deployment flow.